MVP Feature Scope

The most impactful update was the introduction of the new Graph class. This class now contains all graph-related behavior, including adding edges, checking nodes, running Dijkstra, reconstructing paths, and computing total route cost. Centralizing these operations made the rest of the codebase more consistent. Both scheduler.py and scenario.py were rewritten to use graph methods instead of raw dictionary lookups. This forced me to re-evaluate some function designs, but the end result is significantly cleaner.

I also added some docstrings throughout the project, which helps make longer functions in each module easier to read. The scenario parser now has better validation and error messages, which prevent bad input files from crashing the system. The simulation file was improved to track delivery completion more carefully and produce clearer summary statistics. These changes make the project feel more like a real, usable tool rather than a collection of algorithms.

Current Progress

The majority of the system is now implemented. The Graph class is complete and fully integrated. Delivery assignment and all routing strategies work as intended. Scenario parsing and validation are in good shape, and the simulation engine can execute courier routes and track completion times. The remaining tasks are mostly polishing steps: finishing the main.py command-line wrapper, adding a small automated test script, and cleaning up output formatting.

Rewriting the routing logic around the new Graph class took more time than expected, mostly because it required touching multiple modules at once. However, the design is much better now. Dijkstra and path operations feel “owned” by the graph, and other modules only need to call the graph’s methods. The scheduling logic itself was manageable, but getting the 2-swap improvement to behave consistently required a few iterations. On the other hand, adding docstrings and improving scenario parsing were easier than I expected once the module boundaries were clear.

Next Steps

My next priorities are to complete the command-line interface in main.py, add testing files, and polish the simulation output formatting. I also want to do more passes through the docstrings to ensure they are consistent across the project. After that, I’ll focus on making sure all modules run smoothly together and that additional scenario files work without modification.